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College of Computer and Information Sciences
Information Systems Department
Project

Course Title:	Introduction to Databases
Course Code:	IS 220
Course Instructor:	Female: Dr. Hala Alrumaih Male: Mr. Abdulaziz Alyahya
Semester:	1 st Sem. 2024-25
Due Date:	14-10-2024

Instructions:

1. Submission date is on **14-10-2024** via *IMAMU LMS Blackboard* before **5 p.m.**
2. Any attempt to cheat from any of your classmates or from the Internet the whole assignment/project will be marked as **Zero**. Be aware that Blackboard uses plagiarism check tool! (Read and sign **Plagiarism Declaration**)
3. There will be **no extension** in the submission deadline.
4. No **hand writing** is accepted.
5. Use this cover page.

Project Title	Saudi Arabian Railways		
Section No.	372	Group No.	4

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Project Evaluation's Sheet

Criteria	Deserved Points	Criteria's Points
Part # 1: Conceptual Part (CLO 2.2)	5	
• Short description (about one paragraph) of the database application		0.5
• Clear data requirement that describes the entities		0.5
• EER (Entities, Relationship, Attributes, Cardinality, Multiplicity)		3
• Report Formats		1
Part # 2: Logical part	4	
• Map EER model to a relational database schema		2
• Normalize your schema to the third normal form		2
Part # 3: Physical Part (implementation) (CLO 2.1)	6	
Schema Implementation	2.5	
• Use appropriate naming conventions for all of your tables and attributes		0.5
• SQL DDL statements to create database, primary key and foreign keys must be defined		0.5
• Define attributes, data types, constraints e.g. NOT NULL/UNIQUE		0.5
• Explain where and how referential integrity constraints have been incorporated		0.5
• Insert at least 5 rows into each table		0.5
Query implementations	3.5	
• 2 different update queries		0.5
• 2 different delete queries		0.5
• 2 simple select queries related to your tables		0.5
• 2 nested queries related to your tables		0.5
• 2 simple join queries related to your tables		0.5
• 2 simple retrieval queries using group by, having clause, and aggregation functions related to your tables		0.5
• 2 different views, give SQL translations of them, and indicate their implementation and solutions		0.5
Subtotal#1		15
Presentation	5	
• Slides (CLO 3.2)		2.5
• Discussion (CLO 3.1)		2.5
Subtotal#2		5



Client:

The intended client for this database application is the Saudi Arabian Railways (SAR), which is the authorized railway company in KSA. SAR operates and controls a vast rail transport system that links the cities and other parts of the Kingdom. The database system will be implemented and developed primarily to improve the key processes of railway station services including passenger bookings, timetable, and passenger information processing.

Project Description:

The proposed research is intended to come up with an integrated Railway Management System for SAR Railway Station that is an important element of the Saudi railway network. In detailed it will include ticket booking, live train timetable, passenger information and staff administration. Such services shall be automated through the database in order to ensure that passengers at the SAR can access efficient and user-friendly experience from ticket bookings to their travel schedules. The purpose is to ensure the functionality and the commonality and the possibility to improve the state of travelling by providing the convenient interfaces and the up-to-date information.

The database will include the following entities, attributes, and relationships:

Data Requirements:

Entities:

1. Passenger:

- **Attributes:** Passenger_ID, Name, Email, Phone_Number, Date_of_Birth
- **Relationships:**
 - A passenger **books** tickets. (One-to-Many relationship, total participation of Passenger in Ticket)

2. Train:

- **Attributes:** Train_ID, Train_Number, Capacity, Train_Type
- **Relationships:**
 - A train **follows** a route. (Many-to-One relationship, partial participation of Train in Route)



- A train **is assigned** to multiple schedules. (One-to-Many relationship, partial participation of Schedule in Train)

3. Route:

- **Attributes:** Route_ID, Source, Destination, Duration
- **Relationships:**
 - A route **is followed by** multiple trains. (One-to-Many relationship, partial participation of Route in Train)
 - Multiple routes can pass through a station. (Many-to-Many relationship between Route and Station, handled through the association "Passing")

4. Station:

- **Attributes:** Station_ID, Station_Name, Location, Contact_Details
- **Relationships:**
 - A station is managed by multiple employees. (One-to-Many relationship between Station and Employee, total participation of Employee)
 - A station is connected to routes. (Many-to-Many relationship with Route)

5. Schedule:

- **Attributes:** Schedule_ID, Train_ID, Route_ID, Departure_Time, Arrival_Time
- **Relationships:**
 - Each schedule **is assigned** to one train and follows one route. (Many-to-One relationship with Train, total participation of Schedule in Train and Route)

6. Employee:

- **Attributes:** Employee_ID, Name, Position, Shift_Timing
- **Relationships:**
 - Employees **manage** stations. (Many-to-One relationship, total participation of Employee in Station)

7. Ticket:



- **Attributes:** Ticket_ID, Passenger_ID, Train_ID, Booking_Date, Departure_Date, Seat_Number, Price
- **Relationships:**
 - A ticket is **booked by** a passenger. (Many-to-One relationship with Passenger)
 - A ticket is associated with a train. (Many-to-One relationship with Train)
 - **Generalization:** Tickets are divided into two subclasses:
 - Business Class: Has restrictions as an additional attribute.
 - Economy Class: Has amenities as an additional attribute.

8. Subclass - Business Class Ticket:

- **Attributes:** Restrictions
- **Relationships:** Inherits from the Ticket entity and is linked to the Train.

9. Subclass - Economy Class Ticket:

- **Attributes:** Amenities
- **Relationships:** Inherits from the Ticket entity and is linked to the Train.

Participation:

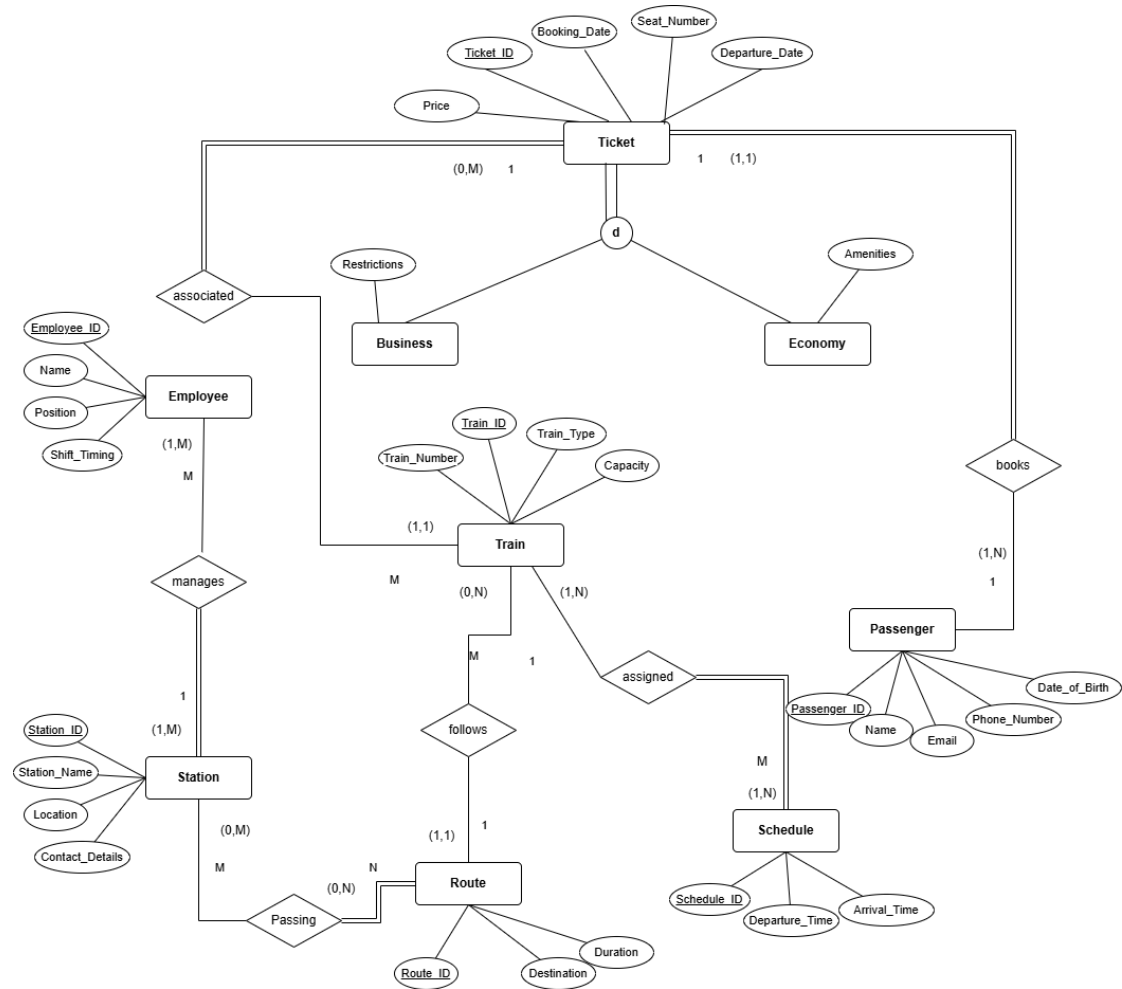
Total Participation:

- **Passenger:** Every passenger must book at least one ticket (partial participation in the Ticket entity).
- **Schedule:** Each schedule must be assigned to a specific train and route (total participation in the Train and Route entities).
- **Employee:** Every employee must be associated with a station (total participation in Station).

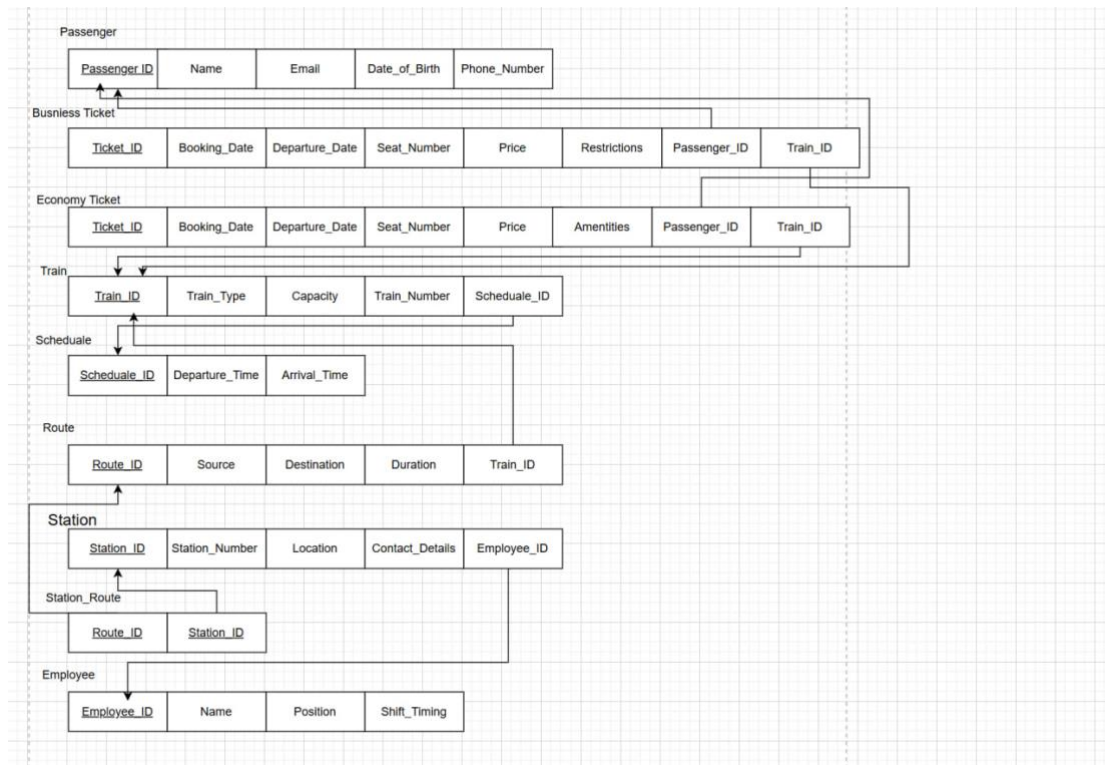
Partial Participation:

- **Train:** Not all trains must necessarily be on a specific route at all times (partial participation in Route).
- **Route:** Not every route will necessarily have a train operating on it at all times (partial participation in Train).

EERD



Map EER model



Normalization

1. Passenger

- **Attributes:** Passenger_ID, Name, Email, Date_of_Birth, Phone_Number
- **Primary Key:** Passenger_ID
- No partial dependencies: All non-key attributes (Name, Email, Date_of_Birth, Phone_Number) are fully dependent on Passenger_ID.
- No transitive dependencies: All attributes depend directly on the primary key.
- **Conclusion:** This table is in 3NF.

2. Business Ticket

- **Attributes:** Ticket_ID, Booking_Date, Departure_Date, Seat_Number, Price, Restrictions, Passenger_ID, Train_ID
- **Primary Key:** Ticket_ID
- No partial dependencies: All non-key attributes (Booking_Date, Departure_Date, Seat_Number, Price, Restrictions, Passenger_ID, Train_ID) depend entirely on Ticket_ID.
- No transitive dependencies: All attributes depend directly on the primary key.
- **Conclusion:** This table is in 3NF.



3. Economy Ticket

- **Attributes:** Ticket_ID, Booking_Date, Departure_Date, Seat_Number, Price, Amenities, Passenger_ID, Train_ID
- **Primary Key:** Ticket_ID
- No partial dependencies: All non-key attributes (Booking_Date, Departure_Date, Seat_Number, Price, Amenities, Passenger_ID, Train_ID) depend entirely on Ticket_ID.
- No transitive dependencies: All attributes depend directly on the primary key.
- **Conclusion:** This table is in 3NF.

4. Train

- **Attributes:** Train_ID, Train_Type, Capacity, Train_Number, Schedule_ID
- **Primary Key:** Train_ID
- No partial dependencies: All non-key attributes (Train_Type, Capacity, Train_Number, Schedule_ID) depend entirely on Ticket_ID.
- No transitive dependencies: All attributes depend directly on the primary key.
- **Conclusion** this table will be in 3NF.

5.Schedule

- **Attributes:** Schedule_ID, Departure_Time, Arrival_Time
- **Primary Key:** Schedule_ID
- No partial dependencies: Both Departure_Time and Arrival_Time are fully dependent on Schedule_ID.
- No transitive dependencies: No other non-key attributes depend on other non-key attributes.
- **Conclusion:** This table is in 3NF.
- .

6. Route

- **Attributes:** Route_ID, Source , Destination ,Duration, Train_ID
- **Primary Key:** Route_ID
- No partial dependencies : All non-key attributes(, Source , Destination ,Duration, Train_ID) depend entirely on: Route_ID
- No transitive dependencies: All attributes depend directly on the primary key.
- **Conclusion:** This table is in 3NF.

7. Station



- **Attributes:** Station_ID, Station_Number, Location, Contact_Details, Employee_ID
- **Primary Key:** Station_ID
- No partial dependencies: All non-key attributes (Station_Number, Location, Contact_Details, Employee_ID) depend entirely on Station_ID.
- No transitive dependencies: the relationship between routes and stations is direct , with no transitive dependence
- **Conclusion:** This table is in 3NF.

8.station_Route

- **Attributes:** Route_ID, Station_ID
- **Primary Key:** (Schedule_ID, Station_ID) (composite Key)
- No partial dependencies: Both : Route_ID and Station_ID from the composite key , and each non-key attribute (if any) would depend on both keys.
- No transitive dependencies: No other non-key attributes depend on other non-key attributes.
- **Conclusion:** This table is in 3NF.

9. Employee

- **Attributes:** Employee_ID, Name, Position, Shift_Timing
- **Primary Key:** Employee_ID
- No partial dependencies: All non-key attributes (Name, Position, Shift_Timing) depend entirely on Employee_ID.
- No transitive dependencies: No non-key attributes are dependent on one another.
- **Conclusion:** This table is in 3NF.

Schema Implementation

Define your database

```
CREATE TABLE Station (  
    Station_ID NUMBER PRIMARY KEY,  
    Station_Name VARCHAR2(100) NOT NULL,  
    Location VARCHAR2(100) NOT NULL,  
    Contact_Details VARCHAR2(100) NOT NULL  
);
```

```
CREATE TABLE Schedule (  
    Schedule_ID NUMBER PRIMARY KEY,  
    Departure_Time TIMESTAMP NOT NULL,  
    Arrival_Time TIMESTAMP NOT NULL  
);
```



```
CREATE TABLE Route (  
    Route_ID NUMBER PRIMARY KEY,  
    Source VARCHAR2(100) NOT NULL,  
    Destination VARCHAR2(100) NOT NULL,  
    Duration NUMBER(3) NOT NULL  
);  
  
CREATE TABLE Train (  
    Train_ID NUMBER PRIMARY KEY,  
    Train_Type VARCHAR2(50) NOT NULL,  
    Capacity NUMBER(4) NOT NULL,  
    Schedule_ID NUMBER,  
    Route_ID NUMBER,  
    FOREIGN KEY (Schedule_ID) REFERENCES Schedule(Schedule_ID),  
    FOREIGN KEY (Route_ID) REFERENCES Route(Route_ID)  
);  
  
CREATE TABLE Route_Station (  
    Route_ID NUMBER,  
    Station_ID NUMBER,  
    PRIMARY KEY (Route_ID, Station_ID),  
    FOREIGN KEY (Route_ID) REFERENCES Route(Route_ID),  
    FOREIGN KEY (Station_ID) REFERENCES Station(Station_ID)  
);  
  
CREATE TABLE Passenger (  
    Passenger_ID NUMBER PRIMARY KEY,  
    Name VARCHAR2(100) NOT NULL,  
    Email VARCHAR2(100) UNIQUE NOT NULL,  
    Date_of_Birth DATE NOT NULL,  
    Phone_Number VARCHAR2(15) UNIQUE NOT NULL  
);  
  
CREATE TABLE Business_Ticket (  
    Ticket_ID NUMBER PRIMARY KEY,  
    Booking_Date DATE NOT NULL,  
    Departure_Date DATE NOT NULL,  
    Seat_Number VARCHAR2(10) NOT NULL,  
    Price NUMBER(8, 2) NOT NULL,  
    Restrictions VARCHAR2(200),  
    Passenger_ID NUMBER,  
    Train_ID NUMBER,  
    FOREIGN KEY (Passenger_ID) REFERENCES Passenger(Passenger_ID),  
    FOREIGN KEY (Train_ID) REFERENCES Train(Train_ID)  
);  
  
CREATE TABLE Economy_Ticket(  
    Ticket_ID NUMBER PRIMARY KEY,
```



```
Booking_Date DATE NOT NULL,  
Departure_Date DATE NOT NULL,  
Seat_Number VARCHAR2(10) NOT NULL,  
Price NUMBER(8, 2) NOT NULL,  
Amenities VARCHAR2(200),  
Passenger_ID NUMBER,  
Train_ID NUMBER,  
FOREIGN KEY (Passenger_ID) REFERENCES Passenger(Passenger_ID),  
FOREIGN KEY (Train_ID) REFERENCES Train(Train_ID)  
);
```

```
CREATE TABLE Employee (  
    Employee_ID NUMBER PRIMARY KEY,  
    Name VARCHAR2(100) NOT NULL,  
    Position VARCHAR2(100) NOT NULL,  
    Shift_Timing VARCHAR2(100) NOT NULL,  
    Station_ID NUMBER,  
    FOREIGN KEY (Station_ID) REFERENCES Station(Station_ID)  
);
```

```
SQL Worksheet  
69 FOREIGN KEY (Passenger_ID) REFERENCES Passenger(Passenger_ID),  
70 FOREIGN KEY (Train_ID) REFERENCES Train(Train_ID)  
71 );  
72  
73 CREATE TABLE Employee (  
74     Employee_ID NUMBER PRIMARY KEY,  
75     Name VARCHAR2(100) NOT NULL,  
76     Position VARCHAR2(100) NOT NULL,  
77     Shift_Timing VARCHAR2(100) NOT NULL,  
78     Station_ID NUMBER,  
79     FOREIGN KEY (Station_ID) REFERENCES Station(Station_ID)  
80 );  
81  
Table created.  
Table created.  
Table created.  
Table created.  
Table created.  
2024 Oracle - Live SQL 24.1.3, running Oracle Database 19c EE Extreme Perf - 19.17.0.0.0 - Database Documentation - Ask Tom - Dev Gym  
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```

Populate your database:

```
INSERT INTO Station (Station_ID, Station_Name, Location, Contact_Details)  
VALUES (1, 'Riyadh Central', 'Riyadh', '011-1234567');
```

```
INSERT INTO Station (Station_ID, Station_Name, Location, Contact_Details)  
VALUES (2, 'Dammam Station', 'Dammam', '013-7654321');
```

```
INSERT INTO Station (Station_ID, Station_Name, Location, Contact_Details)  
VALUES (3, 'Jeddah Station', 'Jeddah', '012-9876543');
```

```
INSERT INTO Station (Station_ID, Station_Name, Location, Contact_Details)  
VALUES (4, 'Makkah Station', 'Makkah', '012-5432198');
```

```
INSERT INTO Station (Station_ID, Station_Name, Location, Contact_Details)  
VALUES (5, 'Madinah Station', 'Madinah', '014-6789102');
```

```
INSERT INTO Schedule (Schedule_ID, Departure_Time, Arrival_Time)
```



```
VALUES (1, TO_TIMESTAMP('2024-10-01 08:00:00', 'YYYY-MM-DD
HH24:MI:SS'), TO_TIMESTAMP('2024-10-01 13:00:00', 'YYYY-MM-DD
HH24:MI:SS'));
INSERT INTO Schedule (Schedule_ID, Departure_Time, Arrival_Time)
VALUES (2, TO_TIMESTAMP('2024-10-02 09:00:00', 'YYYY-MM-DD
HH24:MI:SS'), TO_TIMESTAMP('2024-10-02 11:00:00', 'YYYY-MM-DD
HH24:MI:SS'));
INSERT INTO Schedule (Schedule_ID, Departure_Time, Arrival_Time)
VALUES (3, TO_TIMESTAMP('2024-10-03 07:30:00', 'YYYY-MM-DD
HH24:MI:SS'), TO_TIMESTAMP('2024-10-03 10:30:00', 'YYYY-MM-DD
HH24:MI:SS'));
INSERT INTO Schedule (Schedule_ID, Departure_Time, Arrival_Time)
VALUES (4, TO_TIMESTAMP('2024-10-04 14:00:00', 'YYYY-MM-DD
HH24:MI:SS'), TO_TIMESTAMP('2024-10-04 15:00:00', 'YYYY-MM-DD
HH24:MI:SS'));
INSERT INTO Schedule (Schedule_ID, Departure_Time, Arrival_Time)
VALUES (5, TO_TIMESTAMP('2024-10-05 16:00:00', 'YYYY-MM-DD
HH24:MI:SS'), TO_TIMESTAMP('2024-10-05 20:00:00', 'YYYY-MM-DD
HH24:MI:SS'));
```

```
INSERT INTO Route (Route_ID, Source, Destination, Duration)
VALUES (1, 'Riyadh', 'Dammam', 300);
INSERT INTO Route (Route_ID, Source, Destination, Duration)
VALUES (2, 'Jeddah', 'Makkah', 60);
INSERT INTO Route (Route_ID, Source, Destination, Duration)
VALUES (3, 'Madinah', 'Jeddah', 120);
INSERT INTO Route (Route_ID, Source, Destination, Duration)
VALUES (4, 'Hofuf', 'Al-Ahsa', 75);
INSERT INTO Route (Route_ID, Source, Destination, Duration)
VALUES (5, 'Riyadh', 'Qassim', 240);
```

```
INSERT INTO Train (Train_ID, Train_Type, Capacity, Schedule_ID, Route_ID)
VALUES (1, 'Express', 300, 1, 1);
INSERT INTO Train (Train_ID, Train_Type, Capacity, Schedule_ID, Route_ID)
VALUES (2, 'Regional', 250, 2, 2);
INSERT INTO Train (Train_ID, Train_Type, Capacity, Schedule_ID, Route_ID)
VALUES (3, 'Local', 200, 3, 3);
INSERT INTO Train (Train_ID, Train_Type, Capacity, Schedule_ID, Route_ID)
VALUES (4, 'Freight', 150, 4, 4);
INSERT INTO Train (Train_ID, Train_Type, Capacity, Schedule_ID, Route_ID)
VALUES (5, 'Suburban', 220, 5, 5);
```

```
INSERT INTO Route_Station (Route_ID, Station_ID)
VALUES (1, 1);
INSERT INTO Route_Station (Route_ID, Station_ID)
VALUES (2, 2);
INSERT INTO Route_Station (Route_ID, Station_ID)
VALUES (3, 3);
```



```
INSERT INTO Route_Station (Route_ID, Station_ID)
VALUES (4, 4);
INSERT INTO Route_Station (Route_ID, Station_ID)
VALUES (5, 5);
```

```
INSERT INTO Passenger (Passenger_ID, Name, Email, Date_of_Birth,
Phone_Number)
VALUES (1, 'Ahmed Ali', 'ahmed.ali@gmail.com', TO_DATE('1985-02-12', 'YYYY-
MM-DD'), '0501234567');
INSERT INTO Passenger (Passenger_ID, Name, Email, Date_of_Birth,
Phone_Number)
VALUES (2, 'Sara Al-Faisal', 'sara.faisal@gmail.com', TO_DATE('1990-07-18',
'YYYY-MM-DD'), '0557654321');
INSERT INTO Passenger (Passenger_ID, Name, Email, Date_of_Birth,
Phone_Number)
VALUES (3, 'Omar Zaid', 'omar.zaid@gmail.com', TO_DATE('1995-05-22', 'YYYY-
MM-DD'), '0539876543');
INSERT INTO Passenger (Passenger_ID, Name, Email, Date_of_Birth,
Phone_Number)
VALUES (4, 'Layla Hassan', 'layla.hassan@gmail.com', TO_DATE('1988-11-04',
'YYYY-MM-DD'), '0548765432');
INSERT INTO Passenger (Passenger_ID, Name, Email, Date_of_Birth,
Phone_Number)
VALUES (5, 'Mohammed Al-Harbi', 'mohammed.harbi@gmail.com',
TO_DATE('1979-03-15', 'YYYY-MM-DD'), '0561234987');
```

```
INSERT INTO Business_Ticket (Ticket_ID, Booking_Date, Departure_Date,
Seat_Number, Price, Restrictions, Passenger_ID, Train_ID)
VALUES (101, TO_DATE('2024-10-01', 'YYYY-MM-DD'), TO_DATE('2024-10-
10', 'YYYY-MM-DD'), 'A1', 150.00, 'None', 1, 1);
INSERT INTO Business_Ticket (Ticket_ID, Booking_Date, Departure_Date,
Seat_Number, Price, Restrictions, Passenger_ID, Train_ID)
VALUES (102, TO_DATE('2024-09-25', 'YYYY-MM-DD'), TO_DATE('2024-10-
05', 'YYYY-MM-DD'), 'A2', 140.00, 'None', 2, 2);
INSERT INTO Business_Ticket (Ticket_ID, Booking_Date, Departure_Date,
Seat_Number, Price, Restrictions, Passenger_ID, Train_ID)
VALUES (103, TO_DATE('2024-08-15', 'YYYY-MM-DD'), TO_DATE('2024-08-
20', 'YYYY-MM-DD'), 'A3', 160.00, 'None', 3, 3);
INSERT INTO Business_Ticket (Ticket_ID, Booking_Date, Departure_Date,
Seat_Number, Price, Restrictions, Passenger_ID, Train_ID)
VALUES (104, TO_DATE('2024-07-10', 'YYYY-MM-DD'), TO_DATE('2024-07-
15', 'YYYY-MM-DD'), 'A4', 145.00, 'None', 4, 4);
INSERT INTO Business_Ticket (Ticket_ID, Booking_Date, Departure_Date,
Seat_Number, Price, Restrictions, Passenger_ID, Train_ID)
VALUES (105, TO_DATE('2024-06-05', 'YYYY-MM-DD'), TO_DATE('2024-06-
10', 'YYYY-MM-DD'), 'A5', 155.00, 'None', 5, 5);
INSERT INTO Economy_Ticket (Ticket_ID, Booking_Date, Departure_Date,
Seat_Number, Price, Amenities, Passenger_ID, Train_ID)
```



```
VALUES (201, TO_DATE('2024-10-01', 'YYYY-MM-DD'), TO_DATE('2024-10-10', 'YYYY-MM-DD'), 'B1', 90.00, 'WiFi', 1, 1);
INSERT INTO Economy_Ticket (Ticket_ID, Booking_Date, Departure_Date,
Seat_Number, Price, Amenities, Passenger_ID, Train_ID)
VALUES (202, TO_DATE('2024-09-25', 'YYYY-MM-DD'), TO_DATE('2024-10-05', 'YYYY-MM-DD'), 'B2', 85.00, 'WiFi', 2, 2);
INSERT INTO Economy_Ticket (Ticket_ID, Booking_Date, Departure_Date,
Seat_Number, Price, Amenities, Passenger_ID, Train_ID)
VALUES (203, TO_DATE('2024-08-15', 'YYYY-MM-DD'), TO_DATE('2024-08-20', 'YYYY-MM-DD'), 'B3', 95.00, 'WiFi', 3, 3);
INSERT INTO Economy_Ticket (Ticket_ID, Booking_Date, Departure_Date,
Seat_Number, Price, Amenities, Passenger_ID, Train_ID)
VALUES (204, TO_DATE('2024-07-10', 'YYYY-MM-DD'), TO_DATE('2024-07-15', 'YYYY-MM-DD'), 'B4', 88.00, 'WiFi', 4, 4);
INSERT INTO Economy_Ticket (Ticket_ID, Booking_Date, Departure_Date,
Seat_Number, Price, Amenities, Passenger_ID, Train_ID)
VALUES (205, TO_DATE('2024-06-05', 'YYYY-MM-DD'), TO_DATE('2024-06-10', 'YYYY-MM-DD'), 'B5', 92.00, 'WiFi', 5, 5);
```

```
INSERT INTO Employee (Employee_ID, Name, Position, Shift_Timing, Station_ID)
VALUES (1, 'Ali Saeed', 'Station Manager', 'Morning', 1);
INSERT INTO Employee (Employee_ID, Name, Position, Shift_Timing, Station_ID)
VALUES (2, 'Hassan Ahmed', 'Ticket Officer', 'Evening', 2);
INSERT INTO Employee (Employee_ID, Name, Position, Shift_Timing, Station_ID)
VALUES (3, 'Fatima Al-Shehri', 'Customer Service', 'Morning', 3);
INSERT INTO Employee (Employee_ID, Name, Position, Shift_Timing, Station_ID)
VALUES (4, 'Khalid Omar', 'Maintenance Supervisor', 'Night', 4);
INSERT INTO Employee (Employee_ID, Name, Position, Shift_Timing, Station_ID)
VALUES (5, 'Noura Mohammed', 'Security Officer', 'Evening', 5);
```

```
SQL Worksheet
132
133 INSERT INTO Employee (Employee_ID, Name, Position, Shift_Timing, Station_ID)
134 VALUES (2, 'Hassan Ahmed', 'Ticket Officer', 'Evening', 2);
135
136 INSERT INTO Employee (Employee_ID, Name, Position, Shift_Timing, Station_ID)
137 VALUES (3, 'Fatima Al-Shehri', 'Customer Service', 'Morning', 3);
138
139 INSERT INTO Employee (Employee_ID, Name, Position, Shift_Timing, Station_ID)
140 VALUES (4, 'Khalid Omar', 'Maintenance Supervisor', 'Night', 4);
141
142 INSERT INTO Employee (Employee_ID, Name, Position, Shift_Timing, Station_ID)
143 VALUES (5, 'Noura Mohammed', 'Security Officer', 'Evening', 5);
144

1 row(s) inserted.
1 row(s) inserted.
1 row(s) inserted.
1 row(s) inserted.
1 row(s) inserted.
```

Query implementations

Data Update/Deletion

1. Update Queries

Update Query 1: Update the phone number of a specific passenger



UPDATE Passenger
SET Phone_Number = '0501112233'
WHERE Passenger_ID = 1;

The screenshot shows an Oracle SQL Worksheet interface. The top bar includes 'Clear', 'Find', 'Actions', 'Save', and 'Run' buttons. The SQL editor contains the following code:
1. UPDATE Passenger
2. SET Phone_Number = '0501112233'
3. WHERE Passenger_ID = 1;
4.
Below the editor, the status bar indicates '1 row(s) updated.' The footer text reads: '2024 Oracle - Live SQL 24.1.3, running Oracle Database 19c EE Extreme Perf - 19.17.0.0.0 - Database Documentation - Ask Tom - Dev Gym. Built with using Oracle APEX - Privacy - Terms of Use'.

Update Query 2: Update the ticket price for first-class tickets

UPDATE Business_Ticket
SET Price = 200.00
WHERE Price < 150.00;

The screenshot shows an Oracle SQL Worksheet interface. The top bar includes 'Clear', 'Find', 'Actions', 'Save', and 'Run' buttons. The SQL editor contains the following code:
1. UPDATE Business_Ticket
2. SET Price = 200.00
3. WHERE Price < 150.00;
4.
Below the editor, the status bar indicates '2 row(s) updated.' The footer text reads: '2024 Oracle - Live SQL 24.1.3, running Oracle Database 19c EE Extreme Perf - 19.17.0.0.0 - Database Documentation - Ask Tom - Dev Gym. Built with using Oracle APEX - Privacy - Terms of Use'.

2. Delete Queries

Delete Query 1: Delete a specific train by Train_ID

DELETE FROM Train
WHERE Train_ID = 5;



```
SQL Worksheet
1 DELETE FROM Train
2 WHERE Train_ID = 5;
3
```

1 row(s) deleted.

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Delete Query 2: Delete a passenger with no booking

```
DELETE FROM Passenger
WHERE Passenger_ID NOT IN (SELECT Passenger_ID FROM Business_Ticket
UNION SELECT Passenger_ID FROM Economy_Ticket);
```

```
SQL Worksheet
1 DELETE FROM Passenger
2 WHERE Passenger_ID NOT IN (SELECT Passenger_ID FROM Business_Ticket UNION SELECT Passenger_ID FROM Economy_Ticket);
3
```

1 row(s) deleted.

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II. Data Retrieval (Select) Queries

1. Simple Select Queries

Simple Select Query 1: Retrieve all passengers and their contact information

```
SELECT Name, Phone_Number, Email
FROM Passenger;
```



SQL Worksheet

```
1 SELECT Name, Phone_Number, Email
2 FROM Passenger;
3
```

NAME	PHONE_NUMBER	EMAIL
Ahmed Ali	0501112233	ahmed.ali@gmail.com
Sara Al-Faisal	0557654321	sara.faisal@gmail.com
Omar Zaid	0539876543	omar.zaid@gmail.com
Layla Hassan	0548765432	layla.hassan@gmail.com
Mohammed Al-Harbi	0561234987	mohammed.harbi@gmail.com

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Simple Select Query 2: Retrieve all first-class tickets booked

SELECT Ticket_ID, Seat_Number, Price, Passenger_ID
FROM Business_Ticket;

SQL Worksheet

```
1 SELECT Ticket_ID, Seat_Number, Price, Passenger_ID
2 FROM Business_Ticket;
3
```

TICKET_ID	SEAT_NUMBER	PRICE	PASSENGER_ID
101	A1	150	1
102	A2	200	2
103	A3	160	3
104	A4	200	4
105	A5	155	5

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5 rows selected.

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2. Nested Queries

Nested Query 1: Find the name of passengers who booked tickets for Train 1

SELECT Name
FROM Passenger
WHERE Passenger_ID IN (SELECT Passenger_ID FROM Business_Ticket WHERE
Train_ID = 1);



SQL Worksheet

```
1 SELECT Name
2 FROM Passenger
3 WHERE Passenger_ID IN (SELECT Passenger_ID FROM First_Class_Ticket WHERE Train_ID = 1);
4
```

NAME
Ahmed Ali

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Nested Query 2: Find the trains that have a capacity greater than the average train capacity

```
SELECT Train_ID, Train_Type
FROM Train
WHERE Capacity > (SELECT AVG(Capacity) FROM Train);
```

SQL Worksheet

```
1 SELECT Train_ID, Train_Type
2 FROM Train
3 WHERE Capacity > (SELECT AVG(Capacity) FROM Train);
4
```

TRAIN_ID	TRAIN_TYPE
1	Express
2	Regional

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2 rows selected.

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3. Join Queries

Join Query 1: Retrieve all passengers along with their ticket information

```
SELECT P.Name, T.Ticket_ID, T.Seat_Number, T.Price
FROM Passenger P
JOIN Business_Ticket T ON P.Passenger_ID = T.Passenger_ID;
```



SQL Worksheet

```
1 SELECT P.Name, T.Ticket_ID, T.Seat_Number, T.Price
2 FROM Passenger P
3 JOIN Business_Ticket T ON P.Passenger_ID = T.Passenger_ID;
4
```

NAME	TICKET_ID	SEAT_NUMBER	PRICE
Ahmed Ali	101	A1	150
Sara Al-Faisal	102	A2	200
Omar Zaid	103	A3	160
Layla Hassan	104	A4	200
Mohammed Al-Harbi	105	A5	155

Download CSV

5 rows selected.

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Join Query 2: Retrieve train routes and station names for each route

```
SELECT R.Source, R.Destination, S.Station_Name
FROM Route R
JOIN Route_Station RS ON R.Route_ID = RS.Route_ID
JOIN Station S ON RS.Station_ID = S.Station_ID;
```

SQL Worksheet

```
1 SELECT R.Source, R.Destination, S.Station_Name
2 FROM Route R
3 JOIN Route_Station RS ON R.Route_ID = RS.Route_ID
4 JOIN Station S ON RS.Station_ID = S.Station_ID;
5
```

SOURCE	DESTINATION	STATION_NAME
Jeddah	Makkah	Dammam Station
Madinah	Jeddah	Jeddah Station
Hofuf	Al-Ahsa	Makkah Station
Riyadh	Qassim	Madinah Station

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4 rows selected.

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4. Queries Using Group By, Having Clause, and Aggregation Functions

Query 1: Find the total number of passengers for each train

```
SELECT Train_ID, COUNT(Passenger_ID) AS Total_Passengers
FROM Business_Ticket
GROUP BY Train_ID
HAVING COUNT(Passenger_ID) > 0;
```



SQL Worksheet

```
1. SELECT Train_ID, COUNT(Passenger_ID) AS Total_Passengers
2. FROM First_Class_Ticket
3. GROUP BY Train_ID
4. HAVING COUNT(Passenger_ID) >0;
5.
```

TRAIN_ID	TOTAL_PASSENGERS
1	1
2	1
4	1
5	1
3	1

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Query 2: Retrieve the average ticket price for each train

```
SELECT Train_ID, AVG(Price) AS Avg_Price
FROM Business_Ticket
GROUP BY Train_ID;
```

SQL Worksheet

```
1. SELECT Train_ID, AVG(Price) AS Avg_Price
2. FROM Business_Ticket
3. GROUP BY Train_ID;
4.
```

TRAIN_ID	AVG_PRICE
1	150
2	200
4	200
5	155
3	160

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5 rows selected.

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III. Views

1. View 1: Create a view to show passengers and their first-class ticket details

```
CREATE VIEW Passenger_Tickets_View AS
SELECT P.Name, P.Phone_Number, T.Seat_Number, T.Price
FROM Passenger P
JOIN Business_Ticket T ON P.Passenger_ID = T.Passenger_ID;
SELECT * FROM Passenger_Tickets_View;
```



SQL Worksheet

```
1 CREATE VIEW Passenger_Tickets_View AS
2 SELECT P.Name, P.Phone_Number, T.Seat_Number, T.Price
3 FROM Passenger P
4 JOIN Business_Ticket T ON P.Passenger_ID = T.Passenger_ID;
5 SELECT * FROM Passenger_Tickets_View;
6
```

View created.

NAME	PHONE_NUMBER	SEAT_NUMBER	PRICE
Ahmed Ali	0501234567	A1	150
Sara Al-Faisal	0557654321	A2	200
Omar Zaid	0539876543	A3	160
Layla Hassan	0548765432	A4	200
Mohammed Al-Harbi	0561234987	A5	155

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2. View 2: Create a view to show train schedule details

```
CREATE VIEW Train_Schedule_View AS
SELECT T.Train_ID, T.Train_Type, S.Departure_Time, S.Arrival_Time
FROM Train T
JOIN Schedule S ON T.Schedule_ID = S.Schedule_ID;
SELECT * FROM Train_Schedule_View;
```

SQL Worksheet

```
1 CREATE VIEW Train_Schedule_View AS
2 SELECT T.Train_ID, T.Train_Type, S.Departure_Time, S.Arrival_Time
3 FROM Train T
4 JOIN Schedule S ON T.Schedule_ID = S.Schedule_ID;
5 SELECT * FROM Train_Schedule_View;
6
```

View created.

TRAIN_ID	TRAIN_TYPE	DEPARTURE_TIME	ARRIVAL_TIME
1	Express	01-OCT-24 08:00:00.000000 AM	01-OCT-24 01:00:00.000000 PM
2	Regional	02-OCT-24 09:00:00.000000 AM	02-OCT-24 11:00:00.000000 AM
3	Local	03-OCT-24 07:30:00.000000 AM	03-OCT-24 10:30:00.000000 AM
4	Freight	04-OCT-24 02:00:00.000000 PM	04-OCT-24 03:00:00.000000 PM
5	Suburban	05-OCT-24 04:00:00.000000 PM	05-OCT-24 08:00:00.000000 PM

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